

Modality ceiling

mmlongbench:000014 · chart · 1 gold page

Q	According to the chart on page 14 how much time was spent with family and friends in 2010?		GOLD	21%
T	○	“The document does not provide any evidence or chart on page 14, so it is not possible to determine [...]”		
TL	○	“The document does not provide sufficient textual or numerical information to determine how much [...]”		
V, TLV	✓	“According to the chart on page 14, in 2010, 21% of weekend time was spent with family and friends.”		
NOTE The gold page reaches every rung, so nothing is missing from the context. Only the rungs carrying the page image can read the bar, and the text rungs decline rather than guess.				

Conversion fidelity

mmlongbench:000513 · plain text + table · 1 gold page

Q	How many geographic market names are listed under Los Angeles area?	GOLD	13
T (raw)	✓	“The document lists 13 geographic market names under the Los Angeles area. [...]”	
TL (Paddle)	✗	“26 The document lists 26 geographic market names under the “Los Angeles Area” section [...]”	
TL (MinerU)	✗	“26 The document lists 26 geographic market names under [...]”	
TL (unlimited)	✓	“There are 13 geographic market names listed under the Los Angeles area. [...]”	
TLV	✓	“Based on the provided document, there are 13 geographic market names listed under [...]”	
NOTE The embedded text layer already carried the count; two parsers restructure the list and return its last label instead. The reasoner cannot tell a corrupted number from a real one, and both a better parser and the page image recover it.			

Evidence coverage

mmlongbench:000082 · plain text · 2 gold pages

Q Which method integrates knowledge into white-box models via directive fine-tuning?		GOLD	PKG
oracle	✓	“Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is PKG. [...]”	
drop top	✗	“Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is Fine-tuning Embedding Model. [...]”	
drop bottom	✗	“The method that integrates knowledge into white-box models via directive fine-tuning is directive fine-tuning. [...]”	
NOTE Neither gold page is sufficient alone. Withholding either one does not make the reasoner abstain, it answers from the page that is left, so the missing evidence never announces itself.			

Distractor exposure

mmlongbench:000041 · chart · 1 gold page

Q On the chart showing average exit cap, there is one point that US, Europe, and China have the same average exit cap value. What is this value in USD million? Give me an integer				GOLD	300 million
gold only, +1, +2	✗	“350”			
+3 distractor	✓	“300”			
NOTE The gold page never leaves the context; the added pages only surround it. They change the reading of the same chart rather than removing anything, and this cell is one of the repairs that offset the breaks.					

Evidence integration

mmlongbench:000820 · chart + plain text · 2 gold pages

Q	According to the survey, how many US adults think it's acceptable for the criminal justice system to use automated criminal risk scores? GOLD 1929		
TLV	✗	“According to the survey, 42% of U.S. adults think it is acceptable for the criminal justice system [...]”	
V	✗	“According to the survey, 42% of U.S. adults think it is acceptable for the criminal [...]”	
T	✗	“[...] The survey data (from May 29–June 11, 2018, among 4,594 respondents) is described in detail, including methodology and weighting [...]”	
NOTE The question asks how many, not what percentage. Both operands are read: the chart's 42% at the rungs holding the page image, the 4,594 respondents on the other gold page at T. No rung multiplies them, and 42% of 4,594 is the gold 1,929, so what fails is the join across the two pages and not the evidence on either.			

Response calibration

mmlongbench:000236 · figure · 1 gold page

Q	Among all the pictures, in which page the picture contains the most number of people? Give me the page no. as an integer.	GOLD	9
none	✓	“g”	
abstain	○	“Not answerable”	
NOTE The same page, the same model, and an answer the reasoner is able to give. The abstention instruction turns it into a refusal on a question it can answer.			